Lesson 3.12 - Standard Deviation and Coefficient of Variation

> **Video:** Standard Deviation and Coefficient of Variation by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Formulas

Coefficient of variation (CV): the standard deviation as a percentage of the mean.

$$\mathrm{CV}_{\text{sample}} = \frac{s}{\bar{x}} \times 100\%, \qquad \mathrm{CV}_{\text{population}} = \frac{\sigma}{\mu} \times 100\%$$

Standard deviation:

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}, \qquad \sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

A CV of 20% means the standard deviation is 20% of the mean.

Why use the CV?

- The standard deviation measures how spread out the data is around the mean.
- The CV does the same, but **relative to the mean**, and it has **no units**: in $s/\bar{x}$, feet divided by feet cancel.
- That makes the CV better for **comparing the variability of different data sets**, especially ones with different means or units.

Example - which test has more variation?

Ten students: five took test A and five took test B.

- **Test A:** 79, 84, 86, 81, 89
- **Test B:** 63, 72, 92, 84, 78

Step 1 - means

$$\bar{x} = \frac{\sum x}{n}: \qquad \bar{x}_A = \frac{419}{5} = 83.8, \qquad \bar{x}_B = \frac{389}{5} = 77.8$$

Step 2 - standard deviations

Test A:

x	$x - 83.8$	squared
79	−4.8	23.04
84	0.2	0.04
86	2.2	4.84
81	−2.8	7.84
89	5.2	27.04
Sum		**62.80**

$$s_A = \sqrt{\frac{\sum (x - \bar{x}_A)^2}{n - 1}} = \sqrt{\frac{62.8}{4}} = \sqrt{15.7} \approx 3.962$$

Test B:

x	$x - 77.8$	squared
63	−14.8	219.04
72	−5.8	33.64
92	14.2	201.64
84	6.2	38.44
78	0.2	0.04
Sum		**492.80**

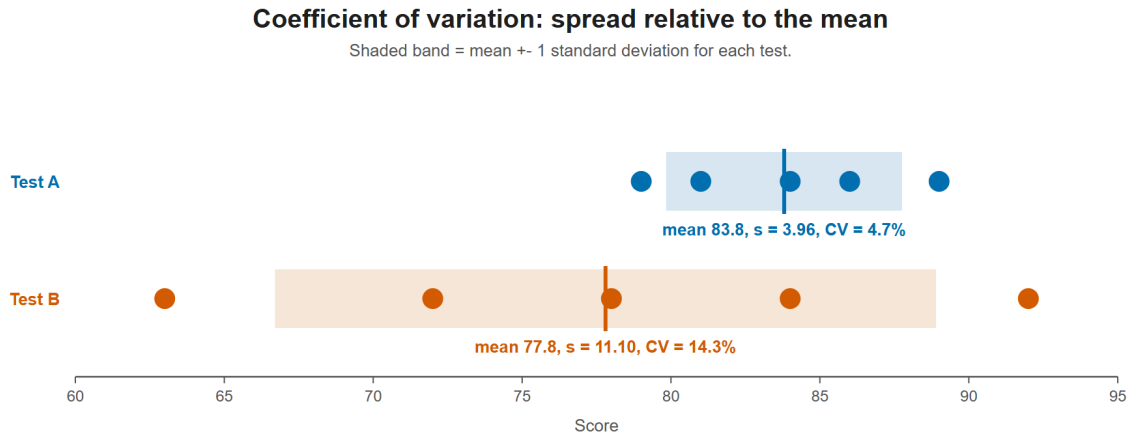
$$s_B = \sqrt{\frac{\sum (x - \bar{x}_B)^2}{n - 1}} = \sqrt{\frac{492.8}{4}} = \sqrt{123.2} \approx 11.10$$

Step 3 - coefficients of variation

$$\mathrm{CV}_A = \frac{s_A}{\bar{x}_A} \times 100\% = \frac{3.962}{83.8} \times 100\% \approx 4.7\%$$

$$\mathrm{CV}_B = \frac{s_B}{\bar{x}_B} \times 100\% = \frac{11.10}{77.8} \times 100\% \approx 14.3\%$$

Conclusion: test B's scores vary much more relative to their mean (14.3% vs 4.7%).



Test B's scores are far more spread out. Its band (mean +- s) is almost three times as wide, and relative to its mean its spread is 14.3% versus 4.7% for test A.

Key takeaways

- CV = SD ÷ mean × 100%.
- The CV is **unitless**, so it is the right tool for comparing the spread of data sets with different scales.
- Calculate the standard deviation first, then divide by the mean.